Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA Certified Network+ Training (N10-009)

TGS Ref No: TGS-2025054472

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 July 2026	Initial release — Network+ (N10-009) 5-day lesson plan aligned to the 36 hands-on labs.	Dr. Alfred Ang
2.0	19 August 2026	Regenerated against the v2 visual slide deck (five N10-009 domains, per-lab folders, TIS web ed-tools: IP Calculator, PCAP Analyzer, Cybersecurity Simulator, Regex Generator). Slide references updated to the new deck.	Dr. Alfred Ang
2.1	19 August 2026	Added runnable lab assets to all 36 labs (80 setup/run/verify/cleanup scripts, 15 service config files, 17 CSV data sets, 36 checklists); new 'What Ships With Every Lab' slide; per-lab 'Files provided with this lab' sections. Slide references rebuilt against the 523-slide deck.	Dr. Alfred Ang

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Course Information

Course Title	CompTIA Certified Network+ Training (N10-009)
WSQ Course Reference	TGS-2025054472
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 8 training hours per day (40 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on the Killercoda Ubuntu Playground and free TIS web tools
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain networking concepts — the OSI model, network appliances, cloud concepts, ports and protocols, traffic types, transmission media, transceivers and connectors, network topologies and IPv4/IPv6 addressing and subnetting. (K1, K2)
- LO2: Implement networks — static and dynamic routing (BGP, EIGRP, OSPF), NAT/PAT, 802.1Q VLANs, trunking, spanning tree, link aggregation, wireless devices and physical installation considerations. (K3, A1, A2)
- LO3: Operate and maintain networks — documentation and IPAM, life-cycle and change management, configuration backup and baselines, SNMP/flow/log monitoring, disaster recovery, DHCP/DNS/NTP network services and secure access methods. (K4, A3, A4)
- LO4: Secure networks — encryption and PKI, IAM/MFA/SSO, RADIUS/TACACS+, physical security, deception technologies, segmentation, common attacks and defence features such as hardening, NAC, ACLs and screened subnets. (K5, A5)
- LO5: Troubleshoot networks — apply the CompTIA troubleshooting methodology with the right tools to cabling, interface, switching, routing, service and performance issues. (K6, A6)

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on networking tasks, 90 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 5 from 4:00 pm (WA 4:00–5:00 pm, PP 5:00–6:30 pm).
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- Courseware and the assessment are on the LMS (https://lms-tms.tertiaryinfotech.com/).

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Course Schedule

The Slides column gives the matching page range in the course deck (CompTIA Network+ (N10-009)-v2.1.pptx) so the trainer can run each session directly from the deck.

Day 1 — Networking Concepts

Time	Duration	Topic / Activity	Slides
9:30–10:00	30 min	Welcome, course introduction, learning outcomes, ground rules and mandatory digital attendance (AM)	—
10:00–11:00	60 min	**Domain 1.0 — Networking Concepts: OSI model layers, encapsulation and PDUs; networking appliances and their functions**	Slides 20–35
11:00–11:15	15 min	Tea break	—
11:15–13:00	105 min	Concepts + hands-on: Lab 1: OSI Model Packet Inspection; Lab 2: IPv4 Subnetting and CIDR	Slides 36–85
13:00–14:00	60 min	Lunch break	—
14:00–15:45	105 min	Concepts + hands-on: Lab 3: IPv6 Addressing and Link-Local; Lab 4: Common Ports and Protocols	Slides 86–106
15:45–16:00	15 min	Tea break	—
16:00–18:15	135 min	Concepts + hands-on: Lab 5: DNS Record Types; Lab 6: DHCP Lease Capture (DORA); Lab 7: Simulating a Cloud VPC with Namespaces	Slides 107–135
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 2 — Network Implementation

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)	—
9:45–11:00	75 min	**Domain 2.0 — Network Implementation: routing technologies (static vs dynamic,**	Slides 136–155

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		BGP/EIGRP/OSPF, route selection, NAT/PAT, FHRP)	
11:00–11:15	15 min	Tea break	—
11:15–13:00	105 min	Concepts + hands-on: Lab 8: Static Routing Between Two Subnets; Lab 9: 802.1Q VLAN Tagging	Slides 156–176
13:00–14:00	60 min	Lunch break	—
14:00–15:45	105 min	Concepts + hands-on: Lab 10: Link Aggregation (LACP Bond); Lab 11: Wireless Concepts and Channel Survey	Slides 177–208
15:45–16:00	15 min	Tea break	—
16:00–18:15	135 min	Concepts + hands-on: Lab 12: NAT and PAT with iptables; Lab 13: DHCP Server with isc-dhcp-server; Lab 14: Authoritative DNS with BIND9	Slides 209–234
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 3 — Network Operations

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)	—
9:45–11:00	75 min	**Domain 3.0 — Network Operations: documentation, IPAM, life-cycle, change and configuration management**	Slides 235–263
11:00–11:15	15 min	Tea break	—
11:15–13:00	105 min	Concepts + hands-on: Lab 15: SNMP Monitoring; Lab 16: NetFlow Analysis with nfdump	Slides 264–276
13:00–14:00	60 min	Lunch break	—
14:00–15:45	105 min	Concepts + hands-on: Lab 17: Centralised Logging with Rsyslog; Lab 18: Time Synchronisation with Chrony	Slides 277–297
15:45–16:00	15 min	Tea break	—
16:00–18:15	135 min	Concepts + hands-on: Lab 19: Configuration Backup with Git; Lab	Slides 298–323

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		20: Performance Baseline with iperf3. Domain 3 recap and consolidation	
18:15–18:30	15 min	Day 3 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 4 — Network Security

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)	—
9:45–11:00	75 min	**Domain 4.0 — Network Security: encryption, PKI, IAM/MFA/SSO, AAA (RADIUS/TACACS+), security terminology and the CIA triad**	Slides 324–337
11:00–11:15	15 min	Tea break	—
11:15–13:00	105 min	Concepts + hands-on: Lab 21: Host Firewall with UFW; Lab 22: TLS Certificate with OpenSSL and Nginx	Slides 338–362
13:00–14:00	60 min	Lunch break	—
14:00–15:45	105 min	Concepts + hands-on: Lab 23: SSH Hardening with Key Authentication; Lab 24: Site-to-Site VPN with WireGuard	Slides 363–375
15:45–16:00	15 min	Tea break	—
16:00–18:15	135 min	Concepts + hands-on: Lab 25: Port Scanning and Service Hardening; Lab 26: Intrusion Detection with Suricata; Lab 27: Building an ACL with iptables	Slides 376–406
18:15–18:30	15 min	Day 4 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 5 — Network Troubleshooting & Assessment

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)	—
9:45–11:00	75 min	**Domain 5.0 — Network Troubleshooting: the CompTIA seven-step methodology; cabling and physical**	Slides 407–434

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		interface issues	
11:00–11:15	15 min	Tea break	—
11:15–13:00	105 min	Concepts + hands-on: Lab 28: Layer 1/2 Diagnostics with ip and ethtool; Lab 29: Reachability with ping, traceroute and mtr; Lab 30: DNS Troubleshooting with dig and resolvectl; Lab 31: ARP and the IPv6 Neighbour Table	Slides 435–468
13:00–14:00	60 min	Lunch break	—
14:00–15:30	90 min	Concepts + hands-on: Lab 32: Packet Capture with tcpdump and TIS PCAP Analyzer; Lab 33: MTU and Path-MTU Discovery; Lab 34: Diagnosing Routing Problems; Lab 35: Simulating Slow and Lossy Links with tc; Lab 36: Service Reachability Triage with curl, nc and openssl. Course recap and exam preparation	Slides 469–506
15:30–15:45	15 min	Tea break	—
15:45–16:00	15 min	**Briefing for Assessment**	—
16:00–17:00	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**	—
17:00–18:30	90 min	**Practical Performance (PP) — hands-on networking tasks, 90 minutes, open book. PM digital attendance**	—

Total training time: 480 minutes (8 hours).

Topic-by-Topic Breakdown

Each domain and lab below shows its starting slide in the course deck.

Domain 1.0 — Networking Concepts (23%) · Slides 20–35

OSI model · appliances · cloud · ports & protocols · traffic types · media · transceivers & connectors · topologies · IPv4 addressing

Lab 1 — OSI Model Packet Inspection · Slide 36

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Objective: Capture live traffic and map each protocol header to its OSI layer.

Activity: Capture HTTP traffic with tcpdump on a Killercoda Ubuntu host, generate requests with curl, and match the MAC, IP, TCP and HTTP headers you see to OSI layers 2, 3, 4 and 7.

Lab 2 — IPv4 Subnetting and CIDR · Slide 77

Objective: Split a /24 into four /26 subnets and verify the math with the TIS IP Calculator.

Activity: Use the TIS IP Calculator to break 192.168.10.0/24 into four /26 subnets, record each network, host range and broadcast, then assign an address from subnet 2 to a Linux interface.

Lab 3 — IPv6 Addressing and Link-Local · Slide 86

Objective: Plan an IPv6 /64, add a global address, and ping link-local using a zone ID.

Activity: Break down 2001:db8::/64 in the TIS IP Calculator's IPv6 mode, add a documentation-range global address to eth0, observe the auto-generated fe80:: link-local address, and ping across both.

Lab 4 — Common Ports and Protocols · Slide 99

Objective: Connect protocol names to well-known port numbers using ss and nmap.

Activity: Enumerate the listening TCP/UDP sockets on your own host with ss, scan the well-known port range of scanme.nmap.org with nmap, and map every open port you find to the N10-009 protocol list.

Lab 5 — DNS Record Types · Slide 107

Objective: Query every N10-009 DNS record type with dig and interpret the answers.

Activity: Use dig to query A, AAAA, MX, TXT, NS, CNAME, PTR, SOA and SRV records against live domains, reading addresses, mail priorities, aliases and zone-authority data from the answers.

Lab 6 — DHCP Lease Capture (DORA) · Slide 115

Objective: Capture all four DHCP DORA messages during a live lease renewal.

Activity: Run tcpdump against UDP ports 67/68 while releasing and re-requesting your DHCP lease with dhclient, then identify the Discover, Offer, Request and Acknowledge messages in the capture.

Lab 7 — Simulating a Cloud VPC with Namespaces · Slide 130

Objective: Build a VPC-style virtual network from namespaces, a bridge and veth pairs.

Activity: Recreate cloud VPC building blocks on one Linux host: two network namespaces act as VMs, a Linux bridge acts as the virtual switch, and veth pairs cable them together on subnet 10.0.1.0/24.

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Domain 2.0 — Network Implementation (20%) · Slides 136–155

Routing technologies · switching technologies · wireless devices · physical installations

Lab 8 — Static Routing Between Two Subnets · Slide 156

Objective: Build a two-subnet topology and route between them with static routes.

Activity: Create three network namespaces (host A, router R1, host B), wire them with veth pairs, enable IP forwarding on R1 and add the static routes that let 10.1.1.0/24 reach 10.2.2.0/24.

Lab 9 — 802.1Q VLAN Tagging · Slide 168

Objective: Create 802.1Q VLAN sub-interfaces and capture a tagged frame.

Activity: Load the 8021q kernel module, create eth0.10 and eth0.20 sub-interfaces each with its own subnet, verify the VLAN IDs with ip -d link, then capture a live tagged frame with tcpdump -e.

Lab 10 — Link Aggregation (LACP Bond) · Slide 177

Objective: Bundle interfaces into an LACP (802.3ad) bond and inspect its state.

Activity: Install ifenslave, load the bonding driver in 802.3ad mode, create bond0, enslave a NIC and read the LACP aggregator, partner MAC and per-slave link state from /proc/net/bonding/bond0.

Lab 11 — Wireless Concepts and Channel Survey · Slide 202

Objective: Survey nearby wireless networks and map channels, bands and security modes.

Activity: Use iw on a wireless laptop (or WiFi Analyzer / Wireless Diagnostics) to scan nearby APs, recording each SSID's channel, signal and security mode, then map channels to bands and compare WEP through WPA3.

Lab 12 — NAT and PAT with iptables · Slide 209

Objective: Configure PAT with iptables MASQUERADE and verify live translations in conntrack.

Activity: Rebuild a namespace client behind the host, enable forwarding, add a MASQUERADE rule on eth0, prove the source address is rewritten with curl ifconfig.me, and inspect the live translation table with conntrack -L.

Lab 13 — DHCP Server with isc-dhcp-server · Slide 219

Objective: Configure isc-dhcp-server to lease addresses on a private subnet.

Activity: Install isc-dhcp-server, create a dummy interface at 10.50.0.1/24, write a scope with range, router, DNS and lease-time options, bind the daemon to dum0 only, start it and inspect the leases file.

Lab 14 — Authoritative DNS with BIND9 · Slide 225

Objective: Host an authoritative DNS zone in BIND9 and query it with dig.

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Activity: Install bind9, write a zone file for lab.local with SOA, NS, A and MX records, register the zone in named.conf.local, validate the syntax with named-checkzone and query your own server with dig @127.0.0.1.

Domain 3.0 — Network Operations (19%) • Slides 235–263

Documentation · life-cycle · change & configuration management · monitoring · DR · network services · access & management

Lab 15 — SNMP Monitoring · Slide 264

Objective: Configure an SNMP agent and poll MIB objects with snmpwalk and snmpget (OBJ 3.2).

Activity: Install snmpd on Ubuntu, configure a read-only v2c community string, then walk the system and IF-MIB subtrees to read uptime, interface names and live traffic counters — the raw data every NMS graphs.

Lab 16 — NetFlow Analysis with nfdump · Slide 270

Objective: Export, collect and analyse flow data to find top talkers and busiest ports (OBJ 3.2).

Activity: Run a flow exporter (fprobe) and collector (nfcapd) on one VM, generate web and ICMP traffic, then use nfdump to build top-talker and destination-port reports from the captured flow records.

Lab 17 — Centralised Logging with Rsyslog · Slide 277

Objective: Build a syslog forwarding pipeline and filter aggregated events by severity and pattern (OBJ 3.2).

Activity: Configure rsyslog to receive on TCP/514 and forward to itself over loopback, inject test events with logger, then verify and filter the aggregated log — including a regex built in the TIS Regex Generator.

Lab 18 — Time Synchronisation with Chrony · Slide 292

Objective: Configure NTP time sync with chrony and interpret stratum, offset and drift (OBJ 3.4).

Activity: Install chrony (the modern ntpd replacement), inspect the selected time sources and stratum hierarchy, add an extra NTP server with iburst, and force an immediate clock step.

Lab 19 — Configuration Backup with Git · Slide 298

Objective: Put network configuration under version control for history, diff, blame and rollback (OBJ 3.1).

Activity: Initialise a git repository in /etc, commit a baseline of the critical network configs, make and commit a change, inspect the audit trail with log and show, then roll the change back with revert.

Lab 20 — Performance Baseline with iperf3 · Slide 305

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Objective: Baseline throughput, jitter and loss with iperf3 TCP, UDP, parallel and reverse tests (OBJ 3.2).

Activity: Run an iperf3 server as a daemon and test against it: a 10-second TCP throughput test, four parallel streams, a 100 Mbps UDP test that reports jitter and loss, and a reverse test for asymmetric paths.

Domain 4.0 — Network Security (14%) • Slides 324–337

Logical & physical security · deception · terminology · audits & compliance · segmentation · attacks · defence features

Lab 21 — Host Firewall with UFW · Slide 338

Objective: Configure a default-deny host firewall with UFW: global allows, source-restricted rules, rule management.

Activity: Set a deny-by-default inbound policy with UFW, permit SSH globally and MySQL only from 10.0.0.0/8, then list, delete and disable rules to manage the firewall lifecycle.

Lab 22 — TLS Certificate with OpenSSL and Nginx · Slide 350

Objective: Generate a self-signed X.509 certificate, serve HTTPS from nginx and inspect the TLS handshake.

Activity: Create an RSA-2048 self-signed certificate with openssl req -x509, configure nginx for HTTPS on 443, then verify the handshake and certificate fields from the client with curl, openssl x509 and s_client.

Lab 23 — SSH Hardening with Key Authentication · Slide 363

Objective: Harden SSH: Ed25519 key authentication, password login disabled, config validated before reload.

Activity: Generate an Ed25519 key pair, install the public key in authorized_keys, prove key login works, then disable PasswordAuthentication in sshd_config, validate with sshd -t and confirm password login is refused.

Lab 24 — Site-to-Site VPN with WireGuard · Slide 368

Objective: Build a site-to-site WireGuard tunnel between two network namespaces and verify the handshake.

Activity: Create two namespaces as VPN endpoints, generate Curve25519 key pairs, configure wg0 interfaces with peer public keys and allowed-ips, bring the tunnel up, ping across it and inspect handshake state with wg show.

Lab 25 — Port Scanning and Service Hardening · Slide 376

Objective: Enumerate exposed services with nmap and reduce the attack surface by disabling what is unneeded.

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Activity: SYN-scan all 65535 TCP ports on your own host, identify service versions and the OS, cross-check each open port against ss -tulnp, disable an unnecessary service and re-scan to confirm the port is closed.

Lab 26 — Intrusion Detection with Suricata · Slide 392

Objective: Run Suricata as an IDS, trigger a known signature and read the alert in fast.log and eve.json.

Activity: Install Suricata, pull the Emerging Threats ruleset, start the engine on eth0, generate benign trigger traffic from testmynids.org, then trace the resulting alert through fast.log and the JSON eve.json event stream.

Lab 27 — Building an ACL with iptables · Slide 398

Objective: Build an ordered iptables ACL that permits MySQL from one subnet and drops it from everywhere else.

Activity: Insert an ACCEPT rule for TCP 3306 from 10.0.0.0/24 above an appended DROP, verify ordering with --line-numbers, test both paths with nc from allowed and blocked sources, then persist the rules across reboots.

Domain 5.0 — Network Troubleshooting (24%) · Slides 407–434

Methodology · cabling & physical interfaces · network services · performance · tools

Lab 28 — Layer 1/2 Diagnostics with ip and ethtool · Slide 435

Objective: Diagnose cable and interface faults from link state, speed/duplex and error counters.

Activity: Use ip and ethtool on a Killercoda Ubuntu VM to read interface state, verify speed and duplex, hunt CRC and frame error counters, and bounce a NIC to recover the link.

Lab 29 — Reachability with ping, traceroute and mtr · Slide 444

Objective: Localise latency and packet loss along a network path with ping, traceroute and mtr.

Activity: Baseline RTT and loss with ping, expose every L3 hop with traceroute, combine both views in a 20-cycle mtr report, then bypass ICMP filters with a TCP traceroute on port 443.

Lab 30 — DNS Troubleshooting with dig and resolvectl · Slide 452

Objective: Isolate DNS faults by comparing resolvers, tracing delegation and reading response codes.

Activity: Identify which resolver the system really uses, compare answers from 1.1.1.1 and 8.8.8.8, walk the delegation chain with dig +trace, time a lookup and flush the systemd-resolved cache.

Lab 31 — ARP and the IPv6 Neighbour Table · Slide 460

Objective: Diagnose L2 adjacency problems with ip neigh and arping, including duplicate IP detection.

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Activity: View and decode neighbour states, probe the default gateway with arping, run duplicate-address detection on your own IP, pin and flush ARP entries, then inspect IPv6 neighbour discovery.

Lab 32 — Packet Capture with tcpdump and TIS PCAP Analyzer · Slide 469

Objective: Capture DNS traffic to a .pcap and analyse it with tshark and the TIS PCAP Analyzer.

Activity: Capture ten DNS packets with a BPF capture filter, dissect them with tshark display filters, then move the .pcap to your laptop and explore it graphically in the browser-based TIS PCAP Analyzer.

Lab 33 — MTU and Path-MTU Discovery · Slide 476

Objective: Demonstrate MTU limits with DF-bit pings and explain PMTUD black holes and MSS clamping.

Activity: Compute the ICMP payload maths (MTU minus 28), send Don't-Fragment pings at and over the limit, lower the interface MTU to 1400 and retest both sides of the new limit, then restore it.

Lab 34 — Diagnosing Routing Problems · Slide 486

Objective: Read and manipulate the routing table; prove which route a destination uses with ip route get.

Activity: Print the routing table, query the kernel's route choice for two destinations, add a static route with a metric, inspect precedence and longest-prefix match, then remove the test route.

Lab 35 — Simulating Slow and Lossy Links with tc · Slide 496

Objective: Reproduce WAN latency, jitter and loss with tc netem and cap bandwidth with tbf.

Activity: Baseline RTT, then use tc netem to inject 200 ms delay, +/-50 ms jitter and 5% loss, verifying each with ping; cap throughput at 1 Mbit/s with tbf, prove it with iperf3, then remove all shaping.

Lab 36 — Service Reachability Triage with curl, nc and openssl · Slide 506

Objective: Isolate a 'site is down' report to TCP, TLS, HTTP or the app using nc, openssl and curl.

Activity: Walk the layered triage: prove the TCP handshake with nc -zv, validate the certificate chain with openssl s_client, read HTTP status codes with curl -v, then break one request into DNS/TCP/TLS/TTFB timings.

Lab Reference (aligned to exam domains)

Exam domain	Weighting	Labs
Domain 1.0: Networking Concepts	23%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6, Lab 7
Domain 2.0: Network Implementation	20%	Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14

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Domain 3.0: Network Operations	19%	Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20
Domain 4.0: Network Security	14%	Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27
Domain 5.0: Network Troubleshooting	24%	Lab 28, Lab 29, Lab 30, Lab 31, Lab 32, Lab 33, Lab 34, Lab 35, Lab 36

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